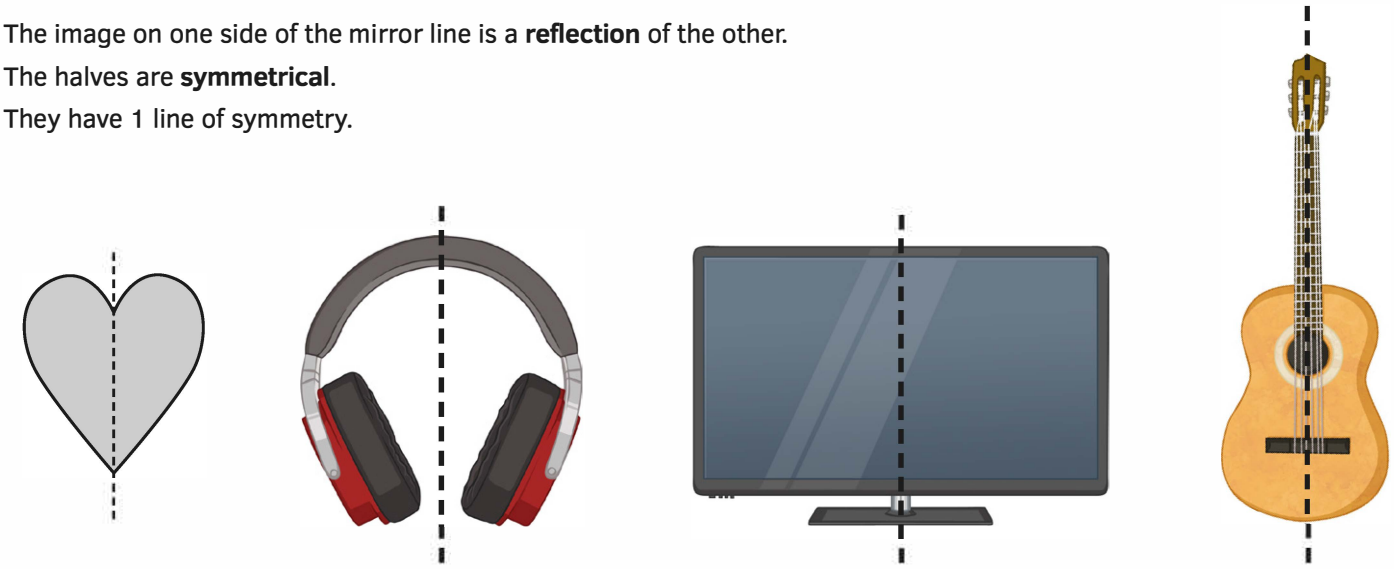
Transformations: **Reflection**

Reflection Symmetry

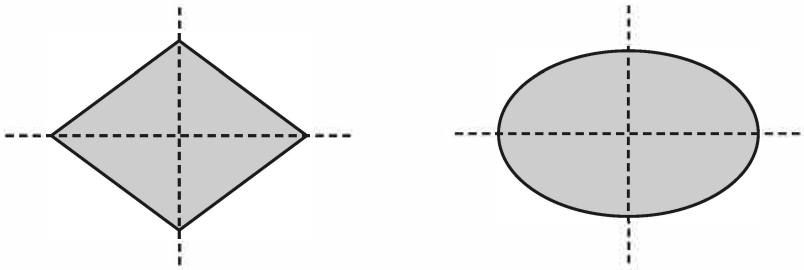
The image on one side of the mirror line is a **reflection** of the other.

The halves are **symmetrical**.

They have 1 line of symmetry.



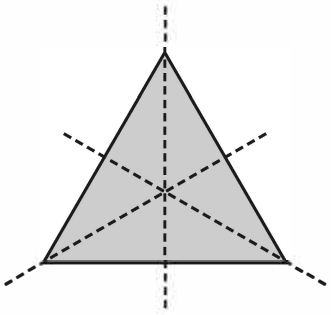
Sometimes it is possible to reflect objects in **two** directions.



These shapes have two lines of symmetry.

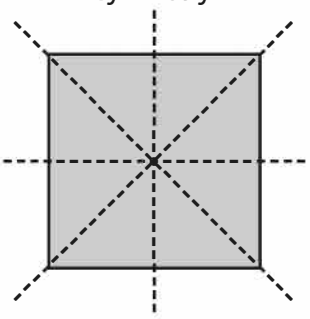
Some shapes have even more!

three lines of symmetry



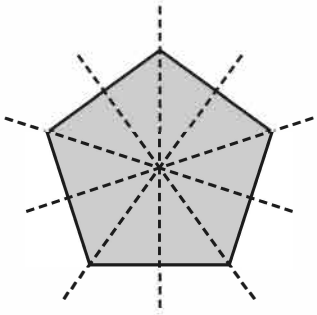
equilateral triangle

four lines of symmetry



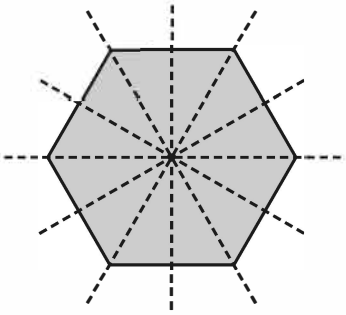
square

five lines of symmetry



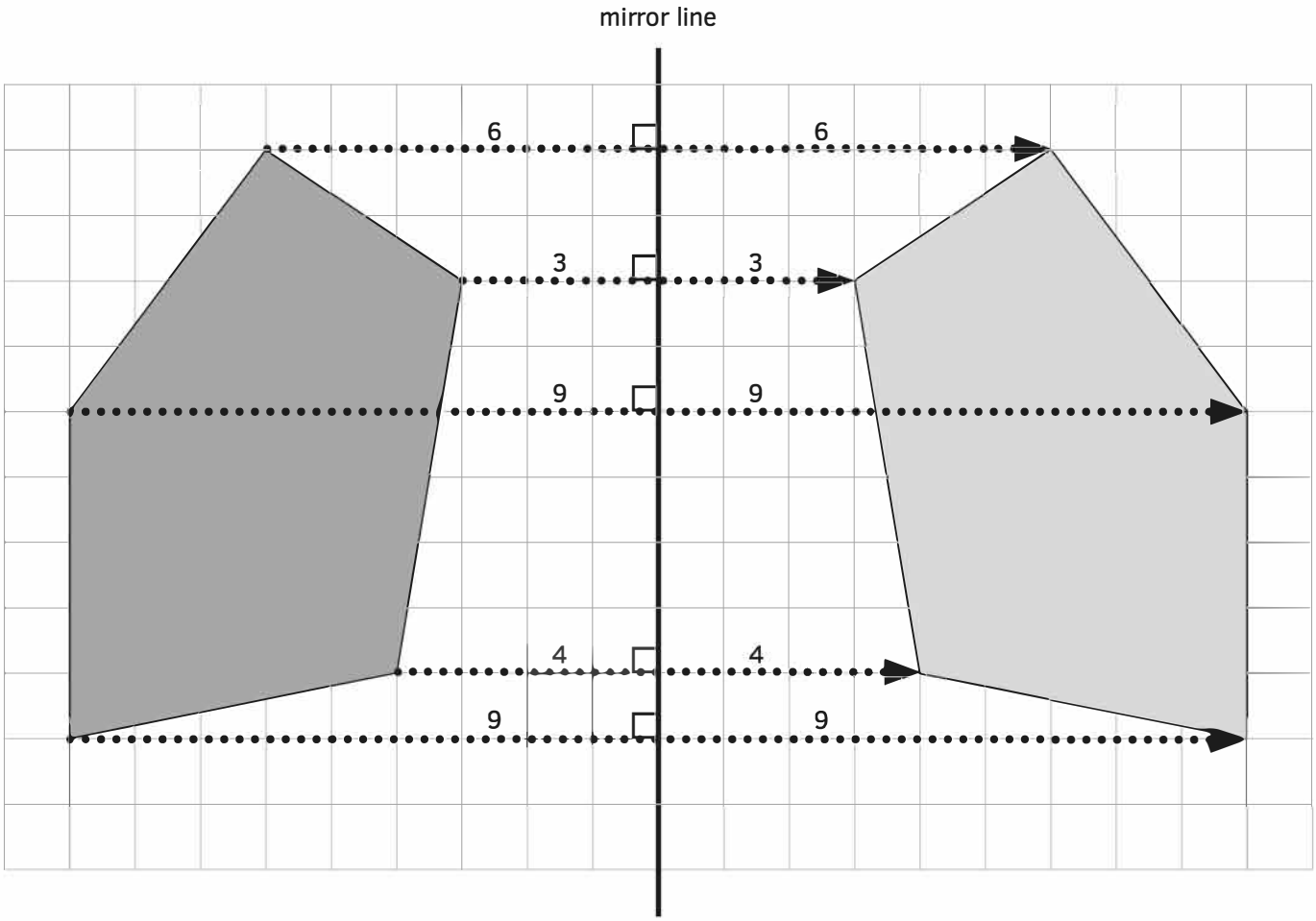
regular pentagon

six lines of symmetry



regular hexagon

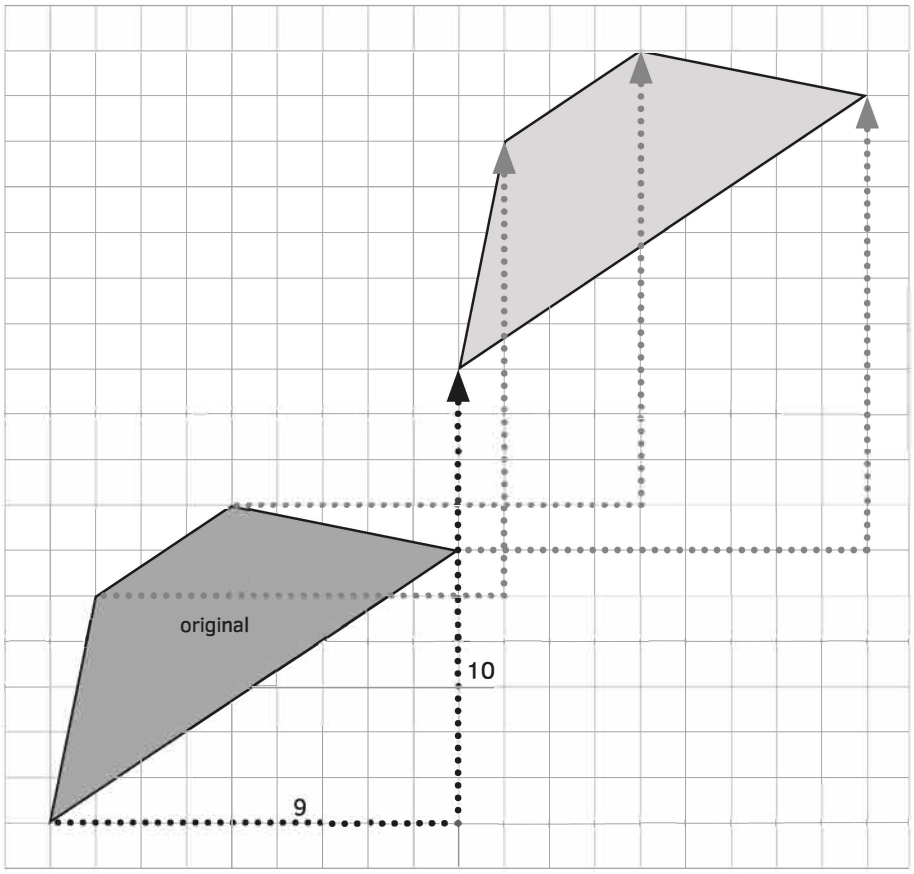
Reflecting Shapes in a Mirror Line



Each point of the original image is the **same distance** from the mirror line as the reflected image.
The line connecting the original point to the reflected point is perpendicular to the mirror line.
The reflected image is **congruent** to the original. This means that the lengths of each side and the angles have not changed.

Transformations: **Translation**

If an object is **translated**, all the points are moved the **same distance** and in the **same direction**.

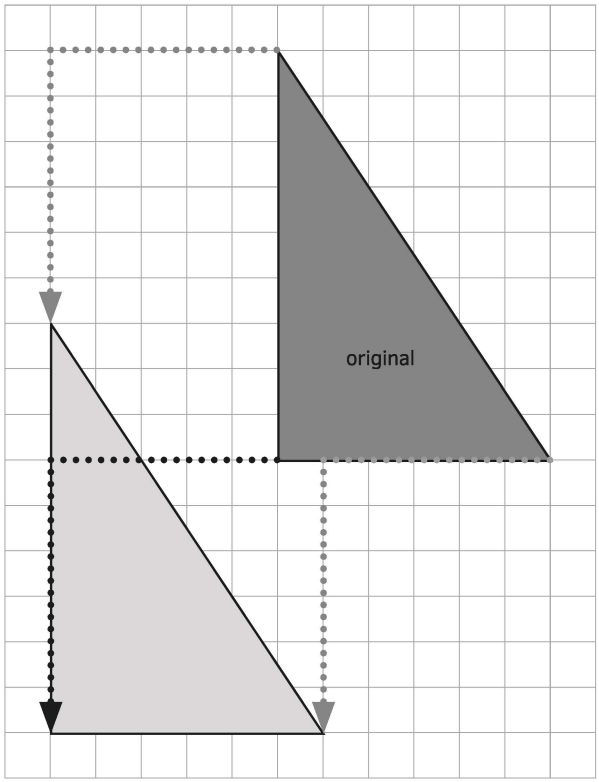


All points have moved 9 → 10 ↑

This translation is $\begin{pmatrix} 9 \\ 10 \end{pmatrix}$

Translations are specified by distance and direction in a VECTOR.

$\begin{pmatrix} \text{Units Right (+) or Left (–)} \\ \text{Units Up (+) or Down (–)} \end{pmatrix}$



The translated image is CONGRUENT to the original. This means that the lengths of the sides and the angles have not changed.

All points have moved 5 ← 6 ↓

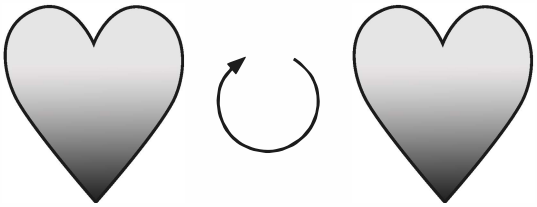
This translation is $\begin{pmatrix} -5 \\ -6 \end{pmatrix}$

Transformations: **Rotation**

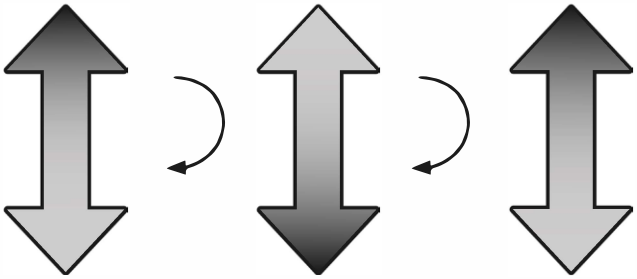
Rotational Symmetry

If an object looks the same after being rotated around its central point, we say it has **rotational symmetry**.
The number of times an object looks the same within a 360° rotation, is called the **order**.

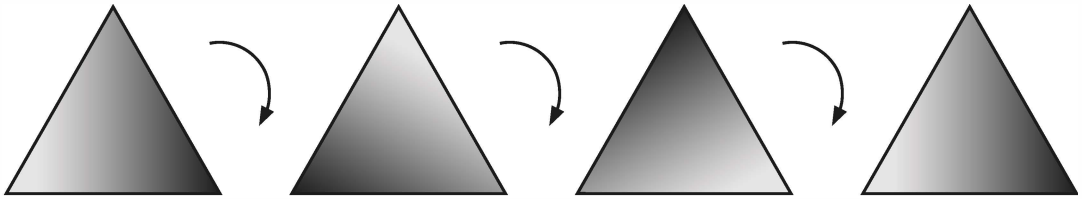
Order 1
This object has to be rotated 360° before it looks the same again.



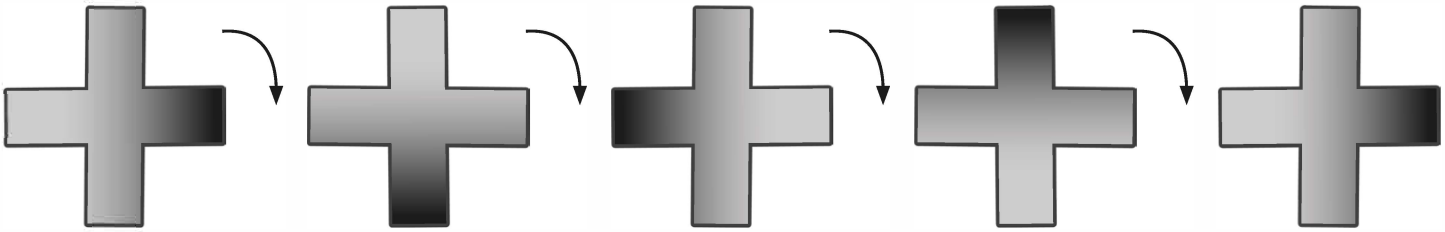
Order 2
This object looks the same **twice** in a 360° rotation.



Order 3
This object looks the same **three** times in a 360° rotation.



Order 4
This object looks the same **four** times in a 360° rotation.



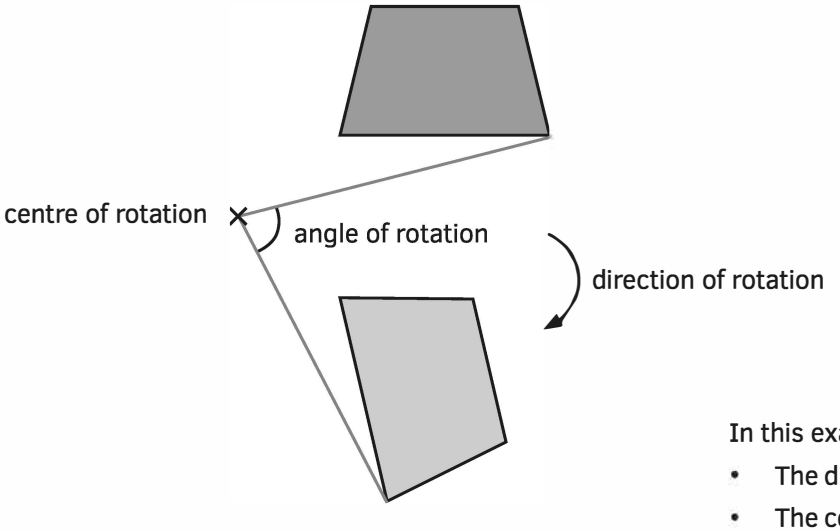
Rotating Shapes

Shapes can be rotated around a fixed point.

The rotated image is **congruent** to the original. This means that the lengths of the sides and the angles have not changed.

Rotations are described by:

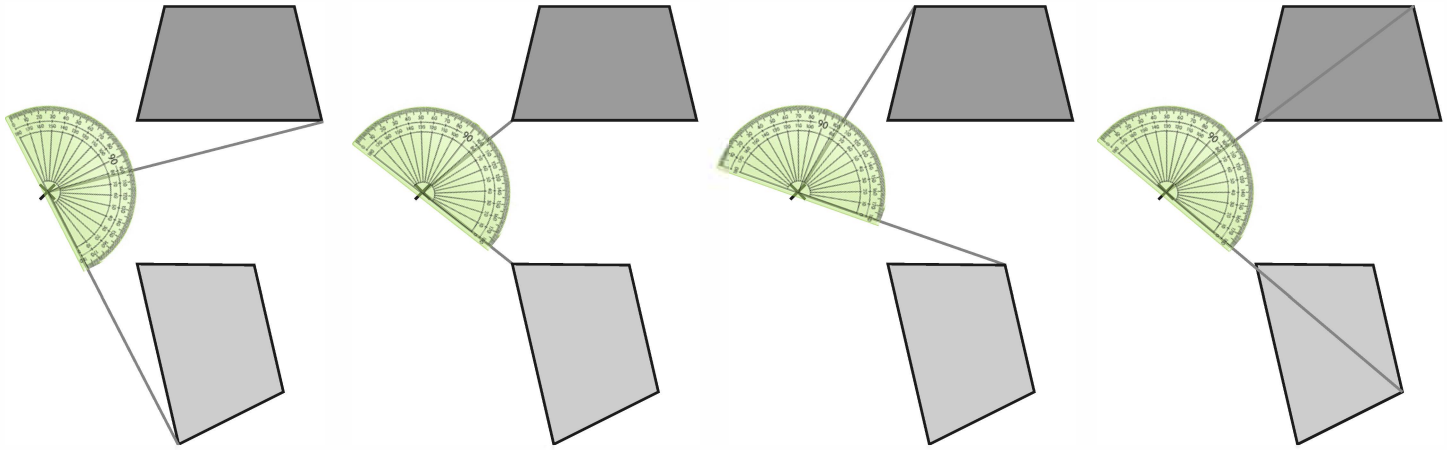
- Direction of rotation – clockwise or anticlockwise?
- Angle of rotation – how much of a turn does **every point** in the object do to create the rotated image?
- Centre of rotation – where are **all** the points rotated from?



In this example:

- The direction of rotation is **clockwise**.
- The centre of rotation has already been given to us (✕).
- What is the angle of rotation?

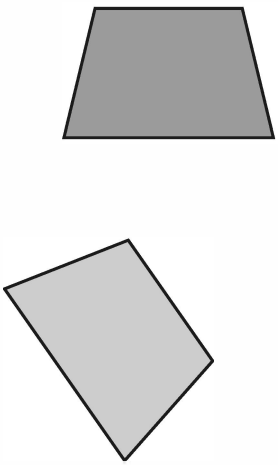
Finding the Angle of Rotation



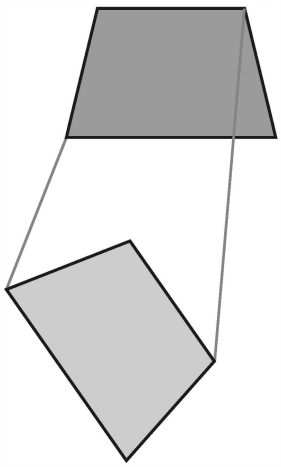
- Every point rotates by the **same distance** and **direction** around the **same point** (centre of rotation).
- By connecting corresponding points to the Centre of Rotation, an angle of turn is created.
- Measuring the turn with a protractor tells us the **angle of rotation**.
- In this example, it is 77°.

Finding the Centre of Rotation

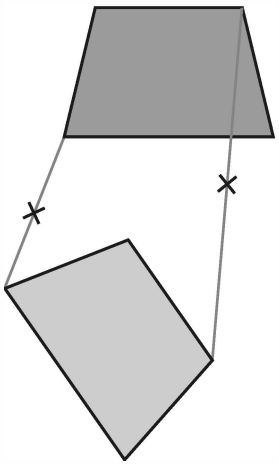
What if we are not given the centre of rotation?



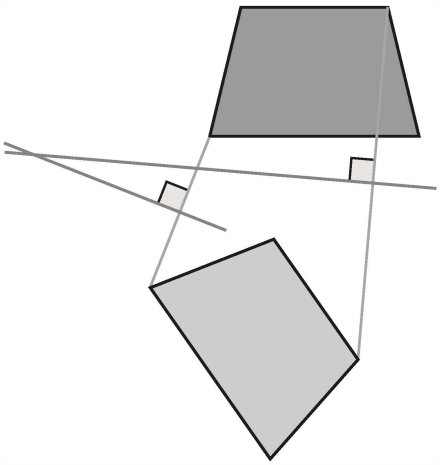
1. Joing 2 corresponding points of your images.



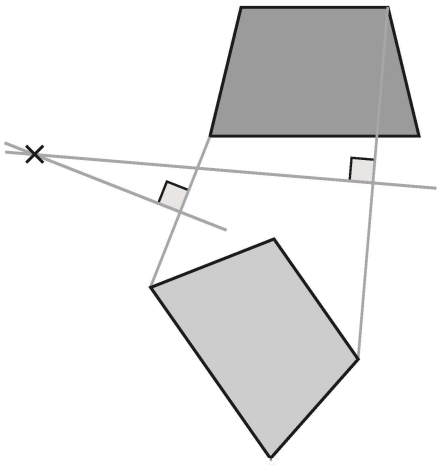
2. Mark the mid point of each line.



3. Draw a perpendicular line through each mid point.



4. The centre of rotation (✕) is where the lines meet.



Transformations: **Enlargement**

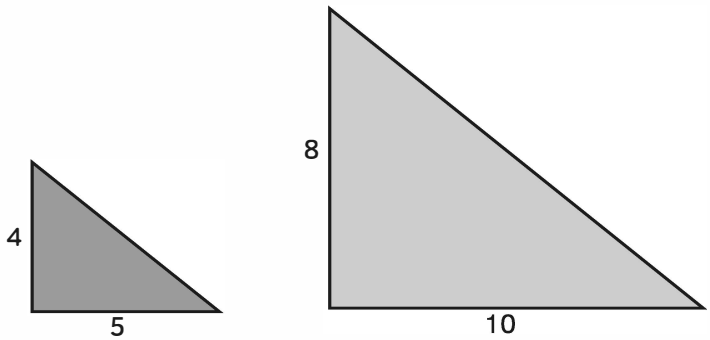
If an object is enlarged, all the edge lengths are **multiplied by the same amount.**

The enlarged image is **similar** to the original. This means that the corresponding angles are the **same** and the corresponding lengths are proportional i.e. all enlarged by the same factor.

An enlarged object can sometimes end up smaller than the original!

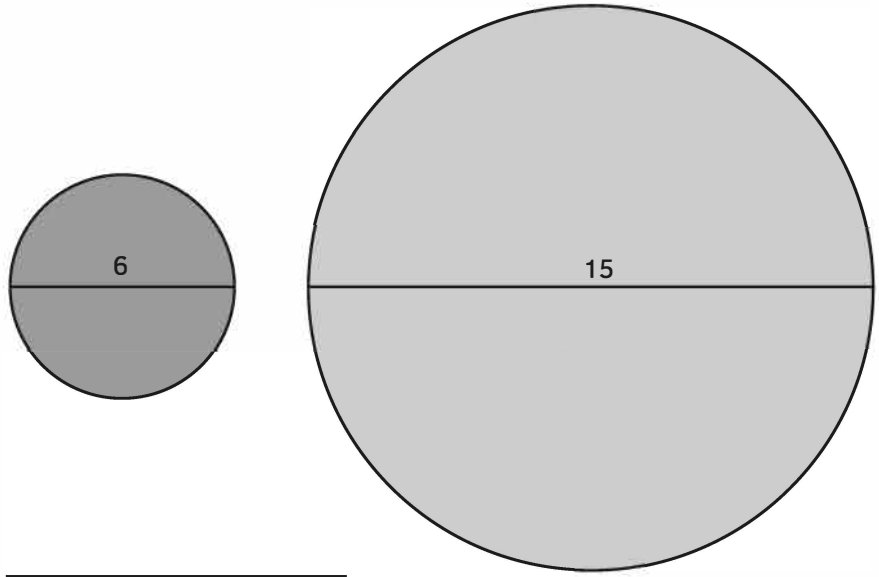
Enlargements are described by:

- **Scale Factor** - how many times the object has been enlarged.



In this example, each side of the triangle has been multiplied by 2.

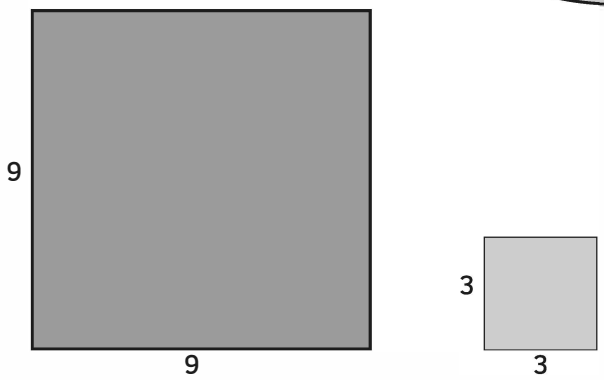
So the object has been enlarged by **scale factor** 2.



In this example, the diameter of the circle has been multiplied by 2.5.

$15 \div 6 = 2.5$

So we say the object has been enlarged by **scale factor** 2.5.



In this example, the sides of the square have been multiplied by $\frac{1}{3}$.

$3 \div 9 = \frac{1}{3}$

So we say the object has been enlarged by scale factor $\frac{1}{3}$.

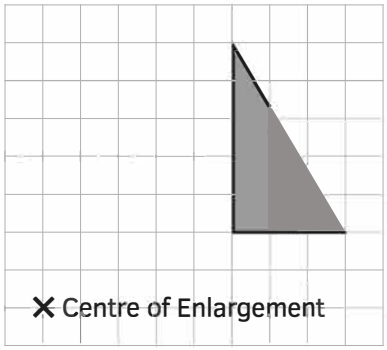
Fraction scale factors make the object smaller.

$$\text{Scale factor} = \frac{\text{length of enlarged side*}}{\text{length of original side*}}$$

*sides **must be** corresponding

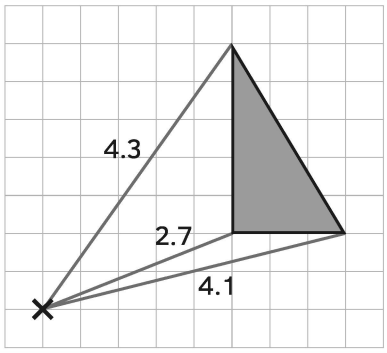
Enlargements can also be described by:

- **Centre of Enlargement** – from where the position of the enlarged shape is measured.
  The distance of each point from the centre of enlargement is multiplied by the Scale Factor to produce the new image.

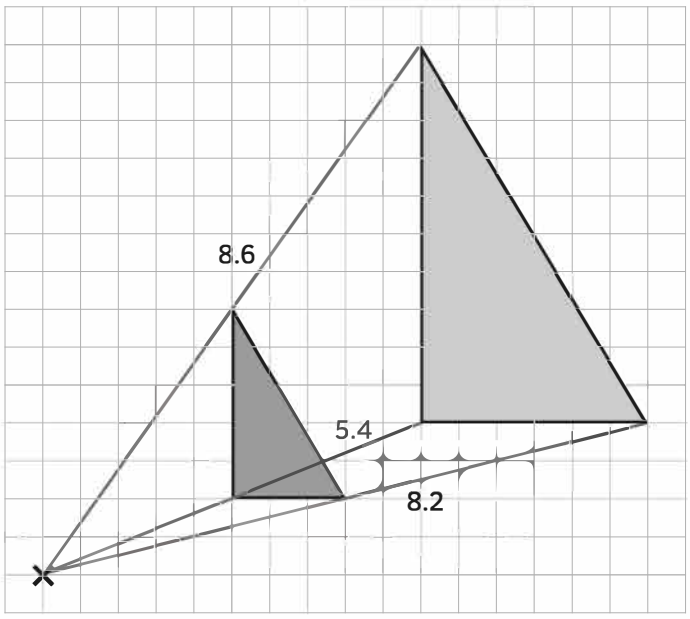


Example:

Enlarge this triangle by Scale Factor 2 from the given centre of enlargement.

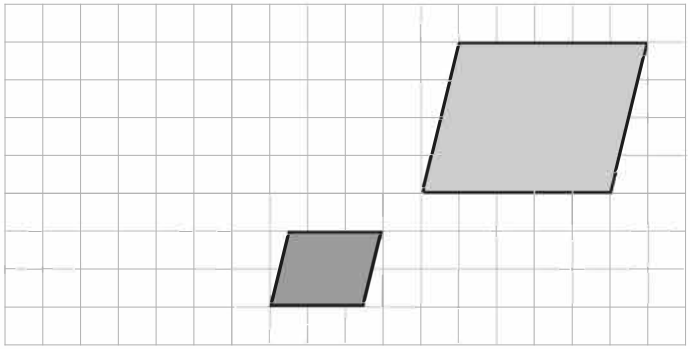


Join each point of the triangle to the centre of enlargement, then measure each line.

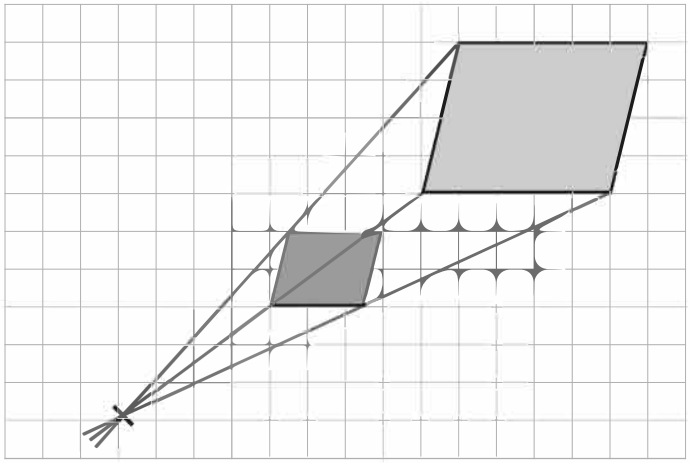


The enlarged object has a scale factor of 2 so the length of each line needs to be multiplied by 2 to produce the points of the new shape.

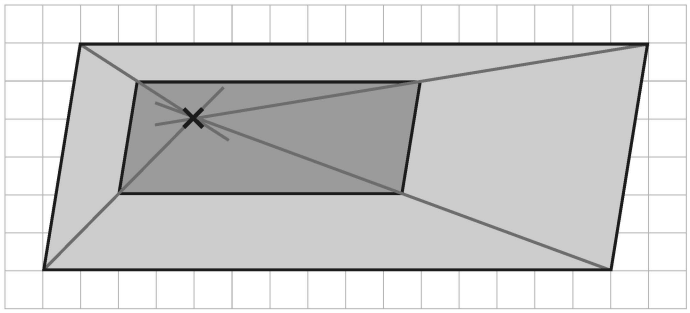
Finding the Centre of Enlargement



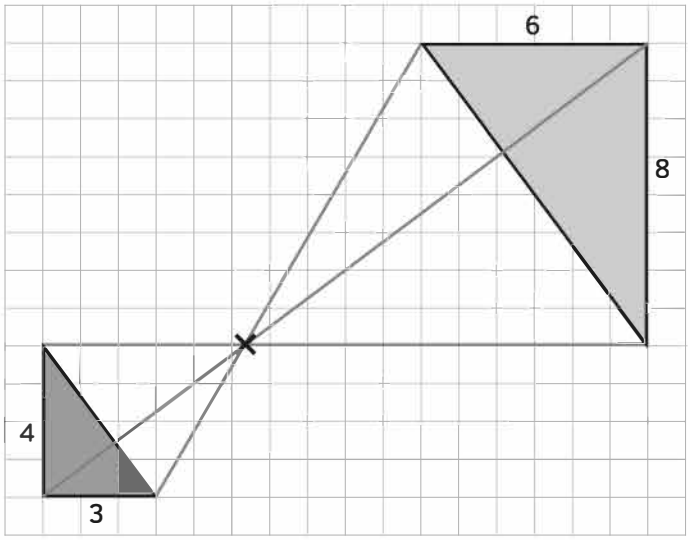
Join the corresponding points of the original and enlarged shape with straight lines.



The Centre of Enlargement is where all the points intercept.



In this example, the Centre of Enlargement is INSIDE the original shape.



In this example, the Centre of Enlargement is **between** the original and enlarged image.

The sides of the triangle have been multiplied by 2, **but** the enlarged object is on the opposite side of the centre of enlargement and rotated by 180°.

So we say the object has been enlarged by scale factor - 2

Negative scale factors mean the object has become enlarged by the given value, but on the opposite side of the centre of enlargement. This will produce a rotated enlargement.